

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8525**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600060
Course Section:	1				

L_29_517_986_1_1505_38427



Name of the candidate. SOHAM SUDESH SHIRKE

Candidate's UID number:

2	0	2	2	6	0	0	0	6	0
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Examination class room number:

4	0	1
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Invigilator's signature with date:

<u>[Signature]</u> 08/10/25

(To be filled in by the candidate)
EXAMINATION: B.E. B. TECH MSE
(For example FY MCA/FY BTECH)

SEMESTER: 7

PROGRAM: CSE (AI & ML)

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: WEDNESDAY 08/10/2025

COURSE NAME: DEEP LEARNING

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Sr. No.

8525



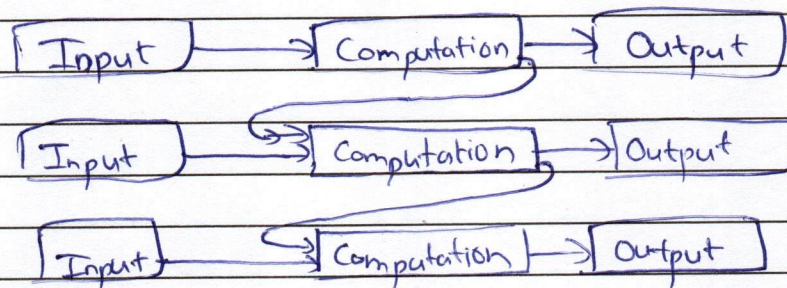
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Q.3A. In Recursive neural network the output of every next input is based on previous computation which is retained by the memory for every computation. The diagram is as follows:



This can work on a small amount of inputs but on facing larger outputs it faces the problem of vanishing gradient.

$$\frac{\partial L}{\partial h_{t-k}} = \text{Value}^{\text{Total inputs}}$$

If th. For eg if the total no. of inputs are 50

$$\frac{\partial L}{\partial h_{t-k}} = \text{Value}^{50}$$

Assume the value is 0.25

$$\frac{\partial L}{\partial h_{t-k}} = 0.25^{50} = 7.889 \times 10^{-31} \text{ which is very close}$$

to zero and the gradient vanishes. This limitation is overcome by LSTM (Long Short Term memory). It has both long term and short term memory so the model does not miss any important detail. It is also a type of RNN but it was designed to overcome vanishing & exploding gradient. Its key components are:

Input layer:

Takes input of data

Three types of gates:

- Forget gate: It determines what information to forget and what is important to pass to the input gate.
- Input gate: It takes input from input layer as well as the forget gate and processes it.
- Output gate: It generates Output.

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MarksQuestion
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As it has long term memory it prevents vanishing gradient and gives more accurate answer. For further preventing vanishing gradient we must use the attention neural network which is used in transformers.

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	Q.3. b)	Gated Recurrent Unit is a type of RNN which helps overcome vanishing and exploding gradient. Its main components are: • Update Gate • Reset Gate Since there are only two gates it is less complex where as in LSTM it has: Input layer: Input of data Three gates: • Forget gate • Input gate • Output gate For faster convergence we must use GRU and it helps in real world applications where speed is more important than generating a proper answer. It is very similar to LSTM & only its computational complexity is less than that of GRU.

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Sr. No. 8525



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